ISE 5405: Optimization I

Lecture 3: Proof Techniques & LaTeX

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The toolkit for everything that follows

What is a Proof?

A proof is a logical argument establishing the truth of a mathematical statement. It is built from:

- **definitions**: precise meanings of the words we use;
- **axioms**: agreed starting points;
- **previous results**: theorems, lemmas, and corollaries;
- **rules of logic**: ways truths combine into new truths.

Course expectation

You will write mathematical arguments throughout the course. A proof is convincing writing for a skeptical but fair reader: your grader.

Logic Warm-up: Implication

Most claims have form $P \Rightarrow Q$, read "if P, then Q."

Statement	Example and status
$P \Rightarrow Q$	If n is even, then n^2 is even.
Contrapositive $\neg Q \Rightarrow \neg P$	If n^2 is odd, then n is odd. Always equivalent to the original.
Converse $Q \Rightarrow P$	If n^2 is even, then n is even. Not automatic; it needs its own proof, even though it is true here.

Negation trap

The negation of "all x satisfy A" is "some x fails A," not "no x satisfies A." Mixing converse and contrapositive is a common error.

The Truth Table for Implication

P	Q	$P \Rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

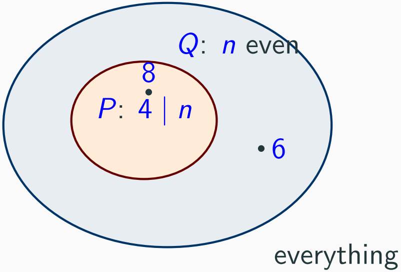
Only one failure

$P \Rightarrow Q$ fails only when P is true and Q is false. A direct proof assumes P and rules out that row by establishing Q.

Vacuous truth

When P is false, the implication makes no violated promise and is true. Empty feasible sets therefore require careful logic, not special arithmetic.

Implication is Containment



$P \Rightarrow Q$ says that the set satisfying P is contained in the set satisfying Q.

The contrapositive reads the same picture from outside: outside Q implies outside P.

The converse swaps the sets. It is false here: 6 is even but not divisible by 4. Converse counterexamples live in the gap.

Technique 1: Direct Proof

Idea

Start from the assumptions and chain valid steps until the conclusion appears.

Claim. If n is even, then n^2 is even.

Proof.

Assume n is even, so $n = 2k$ for some integer k. Then

$$n^2 = (2k)^2 = 4k^2 = 2(2k^2).$$

Since $2k^2$ is an integer, n^2 is twice an integer and therefore even. $\square$

Anatomy

Unpack the definition ($n = 2k$) $\to$ do algebra $\to$ repack the definition. Many direct proofs have exactly this shape.

Technique 2: Proof by Contrapositive

Idea

To prove $P \Rightarrow Q$, prove the equivalent $\neg Q \Rightarrow \neg P$.

Claim. If n^2 is odd, then n is odd.

Proof.

The contrapositive is: if n is even, then n^2 is even. That is precisely the result proved on the previous frame. Hence the original claim holds. $\square$

When to reach for it

Use the contrapositive when $\neg Q$ is easier to manipulate than P. Definitions such as "even $= 2k$" are often friendlier than "odd."

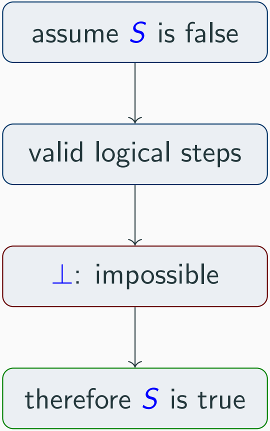
Technique 3: Proof by Contradiction

Idea

Assume the statement is false, derive something impossible, and conclude that the statement must be true.

This is powerful for nonexistence and irrationality claims, where there is nothing to construct.

In LP theory: suppose x^* were not optimal; then a better feasible point would exist; derive a contradiction.



Building the Proof: $\sqrt{2}$ is Irrational

Assume for contradiction that $\sqrt{2} = p/q$, where $p, q \in \mathbb{Z}$, $q \neq 0$, and $\gcd(p, q) = 1$.

1. Squaring gives $p^2 = 2q^2$, so p^2 is even.
2. The lemma "p^2 even $\Rightarrow p$ even" follows by contrapositive: if $p = 2k + 1$, then

   $$p^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1,$$

   which is odd.
3. Write $p = 2k$. Then $4k^2 = 2q^2$, hence $q^2 = 2k^2$, so the same lemma makes q even.
4. Both p and q are even, contradicting $\gcd(p, q) = 1$. Therefore $\sqrt{2}$ is irrational.

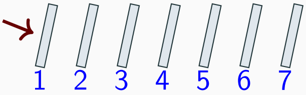
Proof discipline

Do not use the converse "p^2 even $\Rightarrow p$ even" without proving it; the contrapositive supplies exactly the needed lemma.

Technique 4: Proof by Induction

To prove a statement for all $n \in \mathbb{N}$:

1. **Base case**: prove the first case.
2. **Inductive step**: assume the statement at $n = k$ and prove it at $n = k + 1$.



Common bug

The step must work for every relevant k. Do not hide a stronger assumption, such as $k \geq 2$, inside the algebra.

Induction supports correctness proofs for iterative algorithms and recursive structures.

Induction Example: $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$

$$S(n): \qquad \sum_{i=1}^{n} i = \frac{n(n+1)}{2}.$$

Base case:

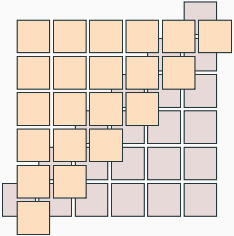
$$1 = \frac{1 \cdot 2}{2}.$$

Assume $\sum_{i=1}^{k} i = k(k+1)/2$. Then

$$\begin{aligned}\sum_{i=1}^{k+1} i &= \sum_{i=1}^{k} i + (k+1) \\ &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{(k+1)(k+2)}{2},\end{aligned}$$

which is the right-hand side of $S(k+1)$.

Why $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$



The sum is a staircase of blocks. A second flipped copy interlocks with it to form an $n \times (n+1)$ rectangle.

$$2(1 + \cdots + n) = n(n+1).$$

Divide by 2. The induction proof verifies the identity; the picture explains why it is true.

Disproving: One Counterexample Suffices

Idea

To disprove a universal claim, exhibit one case where it fails.

Conjecture. For every integer $n \geq 1$, the number $n^2 + n + 41$ is prime.

It holds for $n = 1, 2, \ldots, 39$. Thirty-nine confirmations are evidence, not proof. At $n = 40$,

$$40^2 + 40 + 41 = 1681 = 41^2.$$

Conclusion

The conjecture is false. One verified counterexample is a complete disproof.

The Toolkit at a Glance

Direct: unpack $\rightarrow$ compute $\rightarrow$ repack; use it when a constructive path exists.

Contrapositive: prove $\neg Q \Rightarrow \neg P$; use it when $\neg Q$ is friendlier.

Contradiction: assume the claim is false and reach an impossibility; it is natural for nonexistence, irrationality, and optimality.

Induction: establish a base case and an inductive step for claims indexed by n.

Counterexample: exhibit one verified failing case to disprove a universal claim.

Choosing a technique is itself a skill; practice it on the next frame.

Proof-Technique Trainer

Choose a natural first technique; other correct proofs may exist.

Statement	Natural first tool
$1 + 3 + \cdots + (2n - 1) = n^2$	induction
$\sqrt{3}$ is irrational	contradiction
$n^2 + n + 41$ is prime for every $n \geq 1$	counterexample
If $3n + 2$ is odd, then n is odd	contrapositive
The product of two odd integers is odd	direct proof
Every LP with an optimum has exactly one optimum	counterexample

Selection cue

Look at the claim's logical shape before doing algebra: universal falsehood, nonexistence, implication, or an integer-indexed family.

Proofs Meet LP: a First Optimization Proof

Claim. Let Z_{LP} be the optimal cost of the LP relaxation of a minimization ILP whose optimal cost is Z_{IP}. Then $Z_{\mathrm{LP}} \leq Z_{\mathrm{IP}}$. If an LP optimum x^* is integral, it is ILP-optimal.

Proof.

Every ILP-feasible point is LP-feasible because dropping integrality enlarges the feasible set. Thus $Z_{\mathrm{LP}} \leq Z_{\mathrm{IP}}$.

Now suppose x^* solves the LP and is integral. Then it is ILP-feasible, so

$$Z_{\mathrm{IP}} \leq c^\top x^* = Z_{\mathrm{LP}}.$$

Combining both inequalities yields $Z_{\mathrm{IP}} = Z_{\mathrm{LP}} = c^\top x^*$. □

This short argument is the foundation of branch-and-bound.

Part II — What is LaTeX?

- A plain-text typesetting system for mathematics, proofs, and scientific documents that produces professional PDFs.
- Standard in mathematics, computer science, engineering, and research.
- Strongly recommended for clear homework and research; the syllabus requires one readable PDF, not a particular authoring tool.

Automatic numbering, references, consistent formatting, and precise equations are the payoff.

Source

```
\sum_{i=1}n̂ i = \frac{n(n+1)}{2}
```

Rendered

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}.$$

A minimal source file

```
\documentclass{article}
\usepackage{amsmath,amssymb,amsthm}
\begin{document}
Hello! Inline math: $a2̂+b2̂=c2̂$.
\[
  \min  c^\top x
  \quad\text{s.t.}  Ax\ge b
\]
\end{document}
```

- Preamble: class, packages, and macros.
- Body: text, mathematics, sections, and figures.
- Inline math and displayed math serve different roles.
- A percent sign begins a comment.

Recommended packages include `amsmath`, `amssymb`, `amsfonts`, `bm`, `xcolor`, `graphicx`, `booktabs`, and `hyperref`.

Overleaf: LaTeX in the Browser

- Cloud editor at https://overleaf.com; the free tier is sufficient.
- Real-time collaboration, templates, and version history.
- Start with New Project → Blank Project → write → Recompile.

Keys	Action
Ctrl/Cmd+Enter	Recompile
Ctrl/Cmd+/	Toggle comment
Ctrl+F	Find
Ctrl+L	Go to line

Learn the log

The left panel contains files, logs, and error messages. Reading the first real error is a core LaTeX skill.

Displayed Math: equation vs align

One numbered equation

```
\begin{equation}
  \min  c^\top x    \text{s.t.} Ax\ge b
\end{equation}
```

Aligned lines

```
\begin{align}
  Ax &= b    \label{eq:cons}\\
  x &\ge 0
\end{align}
```

Use `align*` or `\nonumber` to suppress numbering; refer with `\eqref`. Align multi-line derivations at relation symbols. Useful symbols include $\mathbb{R}, \mathbb{Z}, \leq, \geq, \sum, \top, \forall, \exists$.

LaTeX Practice Workflow

1. Start from a small template: LP, induction, matrix, or set notation.
2. Change one symbol or row at a time, then compile.
3. Read the first real error; compare source with rendered mathematics.

Starter LP template

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & c^\top x \\ \text{s.t.} \quad & Ax \le b, \\ & \ell \le x \le u. \end{aligned}$$

Try it

Change $\le$ to $\ge$, add a constraint row with $\backslash\backslash$, and align at `&`. The source templates use `align` and `bmatrix`.

Source pattern

```
\newtheorem{lemma}{Lemma}
\begin{lemma}
If $n$ is even, then $n2̂$ is even.
\end{lemma}
\begin{proof}
Write $n=2k$. Then $n2̂=4k2̂=2(2k2̂)$, so $n2̂$ is even.
\end{proof}
```

The proof environment supplies the QED box. Custom macros save typing:

Macros

```
\newcommand{\R}{\mathbb{R}}
\newcommand{\vect}[1] {\mathbf{#1}}
```

Then `\Rn̂` renders $\mathbb{R}^n$, and `\vect{x}` renders $\mathbf{x}$.

Debugging LaTeX (You Will Need This)

Undefined control sequence: correct a typo or load the needed package.

Missing dollar inserted: put a math-only symbol inside math delimiters.

Runaway argument: find an unmatched brace.

File not found: correct the filename used by `\includegraphics`.

- Read the log and find the first actual LaTeX error.
- Comment out chunks to bisect the problem.
- Temporarily place an early `\end{document}`.
- Build a minimal working example reproducing the failure.

AI help for LaTeX mechanics is allowed under the syllabus; submitted mathematics must still satisfy the course's AI policy.

Wrap-up & Practice

Proof toolkit

- Direct: unpack, compute, repack.
- Contrapositive: flip and negate.
- Contradiction: assume false, demolish.
- Induction: base and step.
- Counterexample: one failure disproves.

Before next class

- Create an Overleaf account and compile a document with an aligned block.
- Prove that the sum of two even integers is even; typeset theorem and proof.
- Revisit the proof-technique examples that were least familiar.

Next

The geometry of polyhedra, where corners become theorems.